

Global cut surgery for tensor-network contraction ordering

Jin-Guo Liu^{1*}

¹ Advanced Materials Thrust, The Hong Kong University of Science and Technology
(Guangzhou), Guangzhou, China

* jinguoliu@hkust-gz.edu.cn

Abstract

Simulated annealing over contraction trees is the strongest general-purpose optimizer of tensor-network contraction orders, yet late in every run its dominant contraction—the tree’s *waist*—freezes. Across 835 instrumented converged states, this frozen waist is *never* a locally minimal balanced cut: a bounded Fiduccia–Mattheyses pass finds a strictly cheaper cut every time, by 3–12 bits. *Global cut surgery* exploits the blind spot: extract the waist, improve it as a graph cut, cold-rebuild both sides, accept on measured cost. It sets three records on ten benchmarks including the Sycamore circuit, holds the high-quality wall of the time–quality Pareto front, and gains with scale; certified bounds and twenty falsified alternatives show search control alone cannot win.

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23 1 Introduction

24 The cost of contracting a tensor network is governed by the order in which its tensors are
 25 combined. Finding a good order is the computational bottleneck of classical random-circuit
 26 simulation [1, 2], of quantum error-correction decoding, and of exact probabilistic inference,
 27 and a rich ecosystem of optimizers has grown around the problem: simulated-annealing tree
 28 refiners [3], hyper-optimized portfolios over greedy and hypergraph-partitioning construc-
 29 tions [4, 5], treewidth machinery from the parameterized-algorithms community [6, 7], exact
 30 methods for small networks [8], and, recently, learned optimizers—reinforcement-learning
 31 policies [9] and LLM-evolved heuristics [10]. Among these, tuned simulated annealing over
 32 contraction trees is in practice the strongest general-purpose refiner: as we show below, thir-
 33 teen mechanically distinct search paradigms and an independent toolchain all converge onto
 34 its frontier, and on small instances that frontier is *certifiably* locally optimal. Improving on it
 35 therefore requires attacking a structural weakness, not a better schedule.

36 This paper identifies such a weakness and builds the algorithm that exploits it. The weak-
 37 ness is the annealer’s *waist*: with the total cost a log-sum-exp over all contractions, the con-
 38 verged tree’s cost is dominated by its single most expensive contraction, which induces a bi-
 39 partition of all tensors. Local tree rewrites move one subtree at a time and can cross between
 40 distinct good bipartitions of the same size only through long, individually uphill move se-
 41 quences, so late in the schedule the waist freezes. Our motivating measurement (Sec. 4) is
 42 that this frozen cut is not merely hard to improve locally—it *is never a locally minimal bal-*
 43 *anced cut of the tensor graph at all.* Across 835 instrumented calls on converged states of
 44 real benchmark networks, a bounded Fiduccia–Mattheyses pass found a strictly cheaper cut of
 45 comparable balance every single time.

46 *Global cut surgery* (Sec. 5) turns that observation into an optimizer: extract the waist
 47 bipartition from the incumbent tree; improve it as a pure graph-cut problem, ignoring the
 48 tree; if a cheaper comparable-balance cut exists, promote it to the tree top, cold-rebuild each
 49 side with fixed-work sub-anneals, splice, and accept only if the independently rescored total
 50 cost drops; iterate on the (possibly moved) waist. An exact rank-non-increasing simplification
 51 pass—which removes up to 89% of the tensors of real-provenance networks before any search
 52 begins—runs as preprocessing.

53 We evaluate the algorithm under a hardened protocol (matched single-threaded budgets,
 54 independent rescoring of every tree from its topology, per-run relabeling, confirmation-based
 55 records; Sec. 2) and re-measure every headline comparison on a single dedicated machine
 56 against the reference Julia implementation, its published benchmark suite, and cotengra (Sec. 6).
 57 Surgery sets records on three of ten instances, its run distribution separates fully from tuned
 58 annealing’s on surface code $d=21$, it holds the entire high-quality wall of the time–quality

Pareto front, and its advantage grows with surface-code distance. Its boundaries are reported with the same care: random 3-regular expanders yield the advantage back at long budgets, and two instances repel it entirely.

Three further results frame the algorithm’s necessity and scope. First, the incumbent must be tuned before being beaten: a single shipped default (an aspirational memory target) costs $20\times$ – $2100\times$, more than any mechanism difference we measured (Sec. 3). Second, on small converged instances nothing can win: we certify the annealing frontier width-optimal, localize the optimum to $[53, 61.5]$ on the circuit proxy, and prove that bounds computable from the isoperimetric profile can go no further (Appendix A)—which is precisely why the gains must come from changing the *input* rather than the search. Twenty falsified alternative mechanisms sharpen the same lesson (Appendix B). Third, on probabilistic-inference workloads the method’s scope splits by structure: annealing-family methods, ours leading, win sparse pedigree networks, while elimination orderings win dense Bayesian networks, and the shipped defaults of TensorInference.jl sit 12–30 bits above the frontier either way (Sec. 7).

2 Setup and evaluation protocol

Instances and cost model. A tensor network is a hypergraph $G = (V, E)$: vertices are tensors and (hyper)edges are indices; contracting to a scalar (or a small open-index set) requires a full binary contraction tree over the tensors. Writing cost_v for the $\log_2$ floating-point cost of internal node v (the sum of $\log_2$ dimensions over the union of the operands’ indices), the two standard quality measures are the total time cost

$$\text{tc} = \log_2 \sum_v 2^{\text{cost}_v}, \quad (1)$$

a log-sum-exp over all $|V|-1$ contractions, and the space cost sc , the $\log_2$ size of the largest intermediate tensor. Minimizing (1) is NP-hard [11], a hardness discovered independently three times: for join ordering in relational databases [12], for sum-of-products loop optimization in compilers [13], and for tensor networks—and algorithms migrate across these communities, most recently join-ordering algorithms into tensor networks [14]. All numbers in this paper are $\log_2$ quantities; a difference of 1.0 is a factor of two in flops. Equation (1) is the root of everything that follows: because tc is a log-sum-exp, the converged tree’s cost is carried almost entirely by its most expensive contraction—the *waist*.

Table 1 lists the ten benchmark instances. Two are closed synthetic networks used for the certification study (a 250-tensor random 3-regular graph and a 561-tensor Sycamore-like circuit proxy, which we distinguish explicitly from the real circuit). The remainder are imported from the OMEinsumContractionOrders benchmark suite [15] with provenance recorded at import: the real Sycamore 53-qubit, 20-cycle network, surface-code syndrome networks at distance $d = 9 \dots 21$, an independent-set counting network on a king graph, a 1000-vertex random 3-regular graph, a 97-qubit random-circuit network, the $n = 28$ queens counting problem, a deep-belief-network inference instance, and a 27-qubit quantum Fourier transform. Section 7 adds ten hard instances from the UAI-2014 inference competition, exported directly from the artifact shipped with TensorInference.jl.

Protocol. Every optimizer run gets 90 seconds of single-threaded wall clock per instance unless stated otherwise. Emitted trees are rescored by an independent checker that recomputes tc and sc from the tree topology and the original index sets alone; optimizer-reported numbers are never trusted. Instances are randomly relabeled per run, so memorized answers keyed on input encoding do not transfer. A best-known value (“record”) only moves when a run

instance	tensors	indices	provenance
reg3_250	250	372	synthetic, closed (certification study)
sycamore_m20	561	963	synthetic Sycamore-like proxy (certification)
sycamore_53_20_0	3369	—	real Sycamore 53q, $m=20$ [4]
surfacecode_d9-d21	219–2203	—	surface-code syndrome networks
ksg	5197	—	king-graph independent-set counting
reg3_1000	1000	1500	random 3-regular, $n=1000$
rqc_97_m24	1238	—	97-qubit random circuit
nqueens_28	4252	—	28-queens counting
dbn_13	572	44 labels	deep-belief-network inference (UAI DBN_13)
qft_27	405	—	27-qubit QFT, 27 open indices

Table 1: Benchmark instances. The first two are the closed synthetic networks of the certification study (Appendix A); the rest are imported real-provenance benchmark networks. Ten additional UAI-2014 inference instances are introduced in Sec. 7.

beats the incumbent by more than 0.05 and a second run under fresh relabeling confirms; the worse of the two runs is recorded. On instances above 1000 tensors, where single-run variance exceeds method differences, scored values are medians of three runs. Per-child CPU-time accounting rejects hidden parallelism. Reference rows (the tuned annealer of Sec. 3, and cotengra’s hyper-optimizer and SA refiner [4]) are measured best-of-three with the identical pipeline.

Same-machine re-measurement. Because the record board accumulated on a development machine, all headline comparisons were re-measured from scratch on a single dedicated 2-core Linux server: 305 scored runs covering budget scaling (90/300/900 s, three repetitions), run distributions (15 repetitions), and the surface-code family (five repetitions), plus a matched-budget ladder of the reference Julia implementation (OMEinsumContractionOrders.jl: TreeSA at both memory-target settings and one/four/eight restarts, HyperND, and three treewidth back-ends) run through that benchmark suite’s own harness, and the inference batch of Sec. 7. Every cross-method statement in Secs. 6 and 7 is same-machine and matched-budget.

3 Tuning the incumbent: a prerequisite

Beating an untuned incumbent is not evidence of algorithmic progress, and the incumbent here ships untuned. The TreeSA-family annealers [3] optimize a composite objective in which the space cost is pulled toward a target sc_{target} , with a penalty active whenever $sc > sc_{\text{target}}$. The implementation we study (omeco, a Rust re-implementation aligned with OMEinsumContractionOrders.jl) inherited the shipped default $sc_{\text{target}} = 20$. On instances whose good trees have naturally larger intermediates—the optimum-quality trees on our two closed instances have $sc = 34$ and 53—this default is not merely unhelpful. It actively drags the search into a low-memory, high-flop region of tree space.

Figure 14 (Appendix C) shows the resulting tc at a fixed 90-second budget as a function of sc_{target} . The transition it reveals is a cliff, not a slope: every target below the natural space cost of good trees is roughly equally harmful (medians 45.6–47.0 on reg3_250, 71–77 on sycamore_m20), quality snaps to the frontier exactly at the natural value ($sc_{\text{target}} = 34$ and 53), and stays there unchanged through $sc_{\text{target}} = \infty$, i.e., with the memory objective deleted. The harm is strictly one-directional. At the shipped default the damage is $4.3 \log_2$ units on reg3_250 and 9.7 on sycamore_m20 ($20\times$ and $800\times$); on the real Sycamore net-

work it reaches 11.1 units ($2100\times$, Sec. 6). The cliff is visible in *published* data as well: the benchmark suite of OMEinsumContractionOrders.jl reports Julia TreeSA at $tc = 71.0$ on the real Sycamore network with its shipped $sc_{\text{target}} = 20$, within 1.6 of our re-measurement of the same configuration, and 11 units above the same annealer with the target removed.

The practical rule is: *never anneal toward a memory target below the natural space cost of good trees; absent a physical memory budget, use the natural value or no target at all.* Every comparison in the remainder of this paper is against the annealer with this fix applied—the *tuned reference*.

Why not better search control. Before targeting the waist, we verified that no search-control change can win where the tuned annealer converges: thirteen mechanically distinct search paradigms and an independent toolchain (cotengra) all land within 0.35 of its frontier on the two closed instances, the frontier is certifiably width-optimal with the optimum localized to $[53, 61.544]$ on the circuit proxy, and we prove that every lower bound computable from the isoperimetric profile is capped below the balanced-cut value. The full certification — the thirteen-mechanism convergence, two sum-form bound theorems, and the impossibility result — is given in Appendix A, and twenty further falsified mechanisms are taxonomized in Appendix B. The lesson they encode is uniform: global information helps only when injected as *input*, never as search control. The remaining question is where such an input exists, and the answer is the waist.

4 The frozen waist is never a minimal cut

Every converged annealed tree has a waist: by (1) its cost is dominated by the argmax-cost contraction, whose leaf set A (against complement B) is a bipartition of all tensors, and whose cost is essentially the weight of the cut (A, B) in the tensor hypergraph. Local tree moves—the associativity rewrites of the annealer [3] (Fig. 3)—transfer one subtree across this bipartition at a time; crossing to a *different* good bipartition of comparable balance requires a long sequence of individually uphill steps, which a cooled annealer will never take; Fig. 1 draws the resulting landscape picture. We also know the freeze is not repairable by small exact windows: exhaustive splicing of all subtrees up to 16 leaves leaves frontier trees unchanged (they are window-exact optimal, Appendix B), while the dominant contraction spans a constant fraction of the network.

The question is whether the frozen waist is actually good as a *cut*. The right tool for that question is Fiduccia–Mattheyses (FM) refinement [16], the classic linear-time partitioning heuristic from circuit design; since it is the one ingredient this paper imports from outside the tensor-network toolbox, we state it in full. Given a bipartition (A, B) of the tensors, assign every tensor a *gain*: the decrease in the cut weight $w(A, B)$ —here the summed $\log_2$ dimensions of the indices straddling the cut—if that tensor alone switched sides. A pass repeatedly moves a highest-gain tensor across the cut *even when that best gain is negative*, locks it, updates the gains of its neighbors, and rejects moves that unbalance the two sides beyond a tolerance; when all tensors are locked, the pass rolls back to the best prefix of its move sequence. The tolerated uphill moves before rollback let a pass carry the boundary across ridges that no sequence of individually improving single moves can cross—the same barriers that freeze the annealer—and a bucket structure over gains makes a full pass linear in the network size. *Bounded* FM caps the number of passes and move attempts (scaling with instance size, nothing tuned per instance), so a call costs a fixed, small slice of the budget. Figure 2 walks one pass on a six-tensor cut.

We instrumented every surgery call of the algorithm of Sec. 5 on its two record runs: at

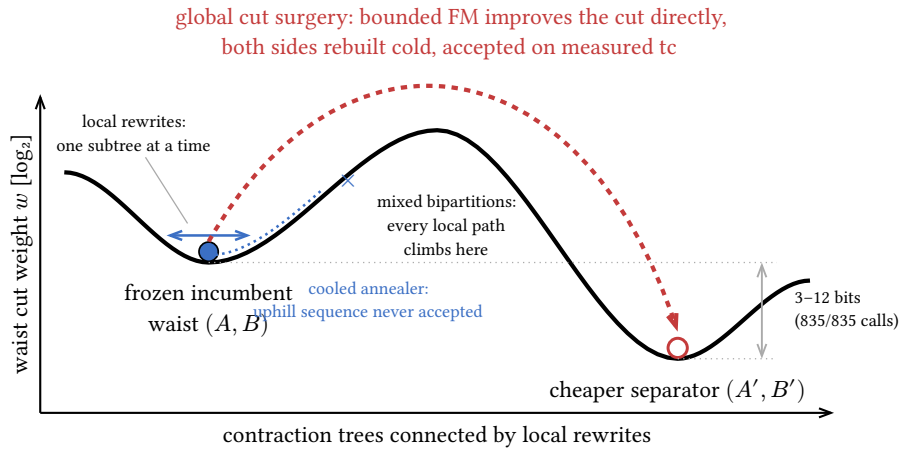


Figure 1: Why the annealer freezes and surgery does not. The landscape is the waist cut weight w over contraction trees, with neighboring trees connected by the annealer’s local rewrites; each rewrite moves one subtree across the waist bipartition, so a distinct separator (A', B') of comparable balance is reachable only through a long sequence of mixed bipartitions of higher cut weight, which a late-schedule annealer never accepts (dotted attempt). Surgery side-steps the barrier by improving the cut in cut space—bounded FM, seeded to jump separators (Sec. 4)—and rebuilding both sides (Algorithm 1, lines 5–8). The well-depth gap is not hypothetical: it is 3–12 bits on every one of the 835 instrumented calls (Fig. 4a). Schematic.

each converged incumbent, extract the waist bipartition and run bounded FM for a cheaper cut of comparable balance on the tensor hypergraph, recording the incumbent’s cut weight, the best alternative found, and the structure of the improved cut.

Figure 4a shows all 835 pooled calls on two instances, and **this panel is the paper’s motivating result**: every single point lies strictly below the diagonal. *The annealer’s waist is never a locally minimal balanced cut*—bounded FM finds a strictly cheaper cut of comparable balance on every call, by margins of 3–12 bits. The improved cuts are sparse rather than near-clique (clique fraction 0.01–0.05): genuine alternative separators, not degenerate rearrangements of the same one. The blind spot is structural and persistent: it does not shrink as the incumbent improves, and it recurs at the new waist after every accepted repair. An optimizer that can see cuts globally therefore always has a move available exactly where the annealer is most stuck. The claim is scoped to the instance families measured here: Sec. 7 exhibits a hub-dominated family on which the balanced waist is sometimes locally minimal, and maps why.

5 The algorithm: global cut surgery

Global cut surgery operationalizes the observation of Sec. 4; Algorithm 1 states the loop, and Fig. 5 walks one iteration on a six-tensor toy network. The optimizer maintains an incumbent contraction tree produced by a standard annealing engine, and interleaves, at every outer iteration, a surgery step that treats the waist as a graph problem:

1. **Extract** (Algorithm 1, lines 3–4). Locate the argmax-cost node of the incumbent (costs recomputed exactly from the index dimensions); its leaf set A versus complement B is the waist bipartition.
2. **Improve the cut, not the tree** (line 5). Run the bounded FM passes of Sec. 4 on the

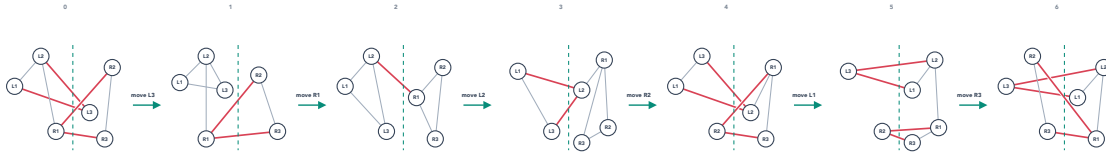


Figure 2: One FM pass on a six-tensor cut. An internal index crosses the cut (dashed) when it connects tensors on both sides; the cut cost is $\sum_{\ell} \log_2 d_{\ell}$ over crossing indices ℓ (red). Each tensor may move at most once per pass: each step moves the not-yet-moved tensor whose transfer most reduces the cut cost while keeping each side within an 18% slack band of its initial size (and nonempty), then locks it; moves may be uphill. The pass ends when every tensor has moved or no legal move remains, and returns the *best cut encountered* along the way.

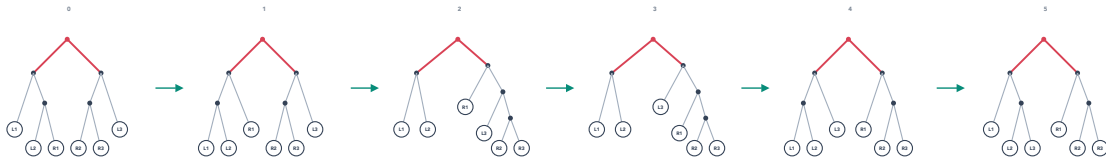


Figure 3: The annealer’s local moves on a six-tensor toy network. Tensors sit at the leaves; the two red branches at the root define the current waist bipartition. Each TreeSA rewrite [3] changes only the association of neighboring subtrees, so moving a tensor to the other side of the waist first requires walking it through the tree—and a global change of the waist requires traversing several intermediate trees, each scored on its own (Fig. 1).

200 tensor hypergraph, with gain defined as the exact reduction in summed $\log_2$ dimensions
 201 of the straddling labels (open output labels are excluded—they are invariant), and bal-
 202 ance constrained to $|A| \pm 18\%$. The search is seeded from the incumbent cut plus four
 203 boundary-BFS alternative bipartitions (one from the highest-degree tensor, three from
 204 random starts), so it can jump to a different separator rather than only polish the current
 205 one.

206 **3. Rebuild cold, with fixed work** (lines 7–8). If a strictly cheaper cut of comparable bal-
 207 ance was found, promote it to the tree top: each side becomes an independent, much
 208 smaller ordering problem (its open indices are the new cut plus its external labels, com-
 209 puted by outside-occurrence counting so the rebuilt tree’s top node matches the inde-
 210 pendent scorer exactly). Anneal each side from scratch with a *fixed-work* budget (three
 211 V-cycles, linear schedules), join, and splice into the incumbent.

212 **4. Accept on measurement** (lines 9–10). Adopt the rebuilt tree only if its independently
 213 recomputed total cost improves; then iterate—the waist typically moves, and surgery
 214 repeats at the new waist until no improving cut is found or the budget ends.

215 Two design choices matter beyond the loop itself. *Fixed work, not fixed time*: giving each
 216 rebuild a deterministic amount of annealing work (rather than a wall-clock slice) collapses
 217 run-to-run spread from 0.8 to 0.08 bits—the algorithm’s variance behavior is set by the rebuild
 218 policy, not by the machine. *Surgery needs its base engine, and vice versa*: an ablation that keeps
 219 only the ratchet base engine plateaus far above the full algorithm (surface-code $d=21$: 49.6
 220 ratchet-only versus 47.38 with surgery; king graph: 50.0 versus 36.8), while surgery without
 221 a competent annealed incumbent has nothing worth repairing. All knobs scale with instance
 222 size; nothing is tuned per instance.

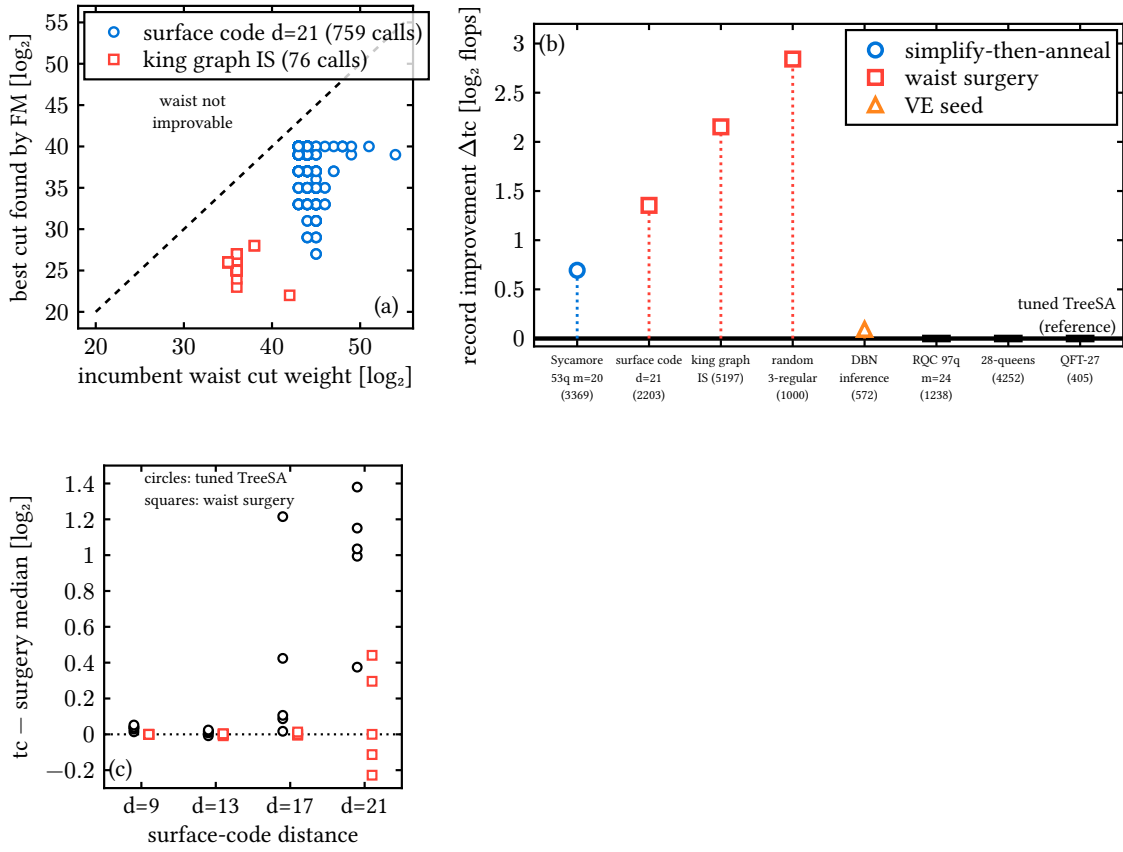


Figure 4: The blind spot and the advantage it buys. (a) Every surgery call on the two record runs: incumbent waist cut weight versus the best comparable-balance cut found by bounded FM. All 835 points lie strictly below the diagonal—the annealed waist is never locally minimal (Sec. 4). (b) Record board: confirmed improvement over the tuned TreeSA reference on each instance (validator protocol: confirm-twice, worse-of-two, per-run relabeling); marker shape encodes the mechanism, thick dashes mark instances where the tuned reference remains the record holder. (c) Surface-code family: all repetitions at 90 s versus code distance, anchored at the surgery median per distance—the surgery advantage grows with d while its run-to-run spread stays collapsed.

223 *Relation to prior refiners.* Refinement layered on top of a construction is an active line:
 224 the TreeSA family itself [3], simulated annealing over partition assignments [17], and, closest
 225 to this work, nearest-neighbor-interchange refinement of hyper-optimizer output, whose gain
 226 over cotengra grows with bond dimension [18]. All of these move the *tree* locally. Surgery
 227 differs in kind: it improves the dominant *cut* as a graph problem and hands the result back
 228 as input to a rebuild—and the measurement of Sec. 4 is precisely that the local refiners’ fixed
 229 point leaves this cut improvable every time.

230 5.1 Preprocessing: exact simplification

231 Real-provenance networks carry large amounts of *rank-non-increasing* structure: tensors whose
 232 contraction with a neighbor can never increase any intermediate rank (rank-1 spiders, dan-
 233 gling legs, series chains). Preprocessing of this kind is established practice (cuTensorNet’s
 234 pathfinder simplifies before partitioning [19], and opt_einsum removes single-tensor reduc-
 235 tions [20]); our pass is the exact rank-non-increasing closure with splice-back, so it composes

Algorithm 1 Global cut surgery. $\mathcal{A}$ is a standard annealing engine (tuned per Sec. 3); $w(A, B)$ is the summed $\log_2$ dimension of the labels straddling the bipartition (A, B) , excluding open output labels. All costs are recomputed exactly from the index dimensions; nothing is tuned per instance.

Require: tensor network G (after the exact simplification pass of Sec. 5.1), time budget

```

1:  $T \leftarrow \mathcal{A}(G)$  ▷ incumbent contraction tree
2: while budget remains do
3:    $v^* \leftarrow \arg \max_v \text{cost}_v(T)$  ▷ the waist: argmax-cost contraction
4:    $(A, B) \leftarrow$  leaf bipartition of  $T$  induced by  $v^*$ 
5:    $(A', B') \leftarrow \text{BOUNDED FM}(G, (A, B))$  ▷ gain =  $w$  reduction; balance  $|A| \pm 18\%$ ;  
seeded from  $(A, B)$  plus four boundary-BFS bipartitions
6:   if  $w(A', B') < w(A, B)$  then
7:      $T_L \leftarrow \mathcal{A}(G[A']), T_R \leftarrow \mathcal{A}(G[B'])$  ▷ cold rebuilds, fixed work (three V-cycles);  
open indices by outside-occurrence counting
8:      $T' \leftarrow \text{SPLICE}(T_L, T_R)$ 
9:     if  $\text{tc}(T') < \text{tc}(T)$  by independent rescoring then
10:       $T \leftarrow T'$  ▷ waist moves; next pass repairs the new waist
11:     end if
12:   else
13:     run annealing sweeps of  $\mathcal{A}$  on  $T$  ▷ interleaved local refinement
14:   end if
15: end while
16: return  $T$  ▷ emitted eagerly throughout (anytime)

```

236 with the record protocol. Before any search, the pass contracts this structure away greedily;
 237 the optimizer runs on the reduced network, and the contracted fragments are spliced back into
 238 the final tree as fixed subtrees (Fig. 6). The pass is exact and costs milliseconds.

239 Its effect is largest exactly where synthetic benchmarks never look (Fig. 15a, Appendix C):
 240 88.7% of the real Sycamore network’s 3369 tensors are removed before annealing begins
 241 ($3369 \rightarrow 381$), 65.5% of surface-code $d=21$, 33.9% of nqueens_28—and 0% of the random
 242 3-regular control, which is why the identical mechanism, correctly measured as useless on
 243 the pre-simplified synthetic proxy two cycles earlier, was nearly discarded. A matched-budget
 244 ablation on the real Sycamore network isolates the contribution: the same anneal reaches
 245 $\text{tc} = 61.3$ with simplification and 76.0 without it, a $2^{14.7}$ ($\sim 27000\times$) difference from prepro-
 246 cessing alone. Figure 15b shows the resulting record march: three successive simplify-then-
 247 anneal implementations drove the Sycamore record $60.605 \rightarrow 60.169 \rightarrow 60.104 \rightarrow 59.911$.

248 5.2 Seeding and anytime behavior

249 The pipeline emits its incumbent eagerly (atomic writes), so it is an anytime optimizer, and
 250 its construction stage is pluggable. Two components earn their place. *Elimination seeds*: on
 251 hypergraph-like instances with few, high-degree labels, seeding the annealer with a min-fill
 252 variable-elimination order gives it a start ($\text{tc} = 28.8$ at 0.2 s on the DBN instance) that greedy
 253 construction (44.7) cannot approach. During the campaign this mechanism appeared to have
 254 a sharp boundary—at 4086 labels the seed seemed to freeze—but the freeze later proved
 255 to be largely an implementation artifact of the greedy construction stage, and a quotient-
 256 graph reimplementations removes it entirely (Sec. 7, “a seed that scales”). *Racing*: arms with
 257 budget-fraction linear schedules, composed by a router, produced three records in a single
 258 run during development; a probe-leader policy never won. Harness-clocked time-to-frontier
 259 records exist as well (16.1 s to the reg3_250 frontier versus 20.2 s for the reference; 0.2 s on

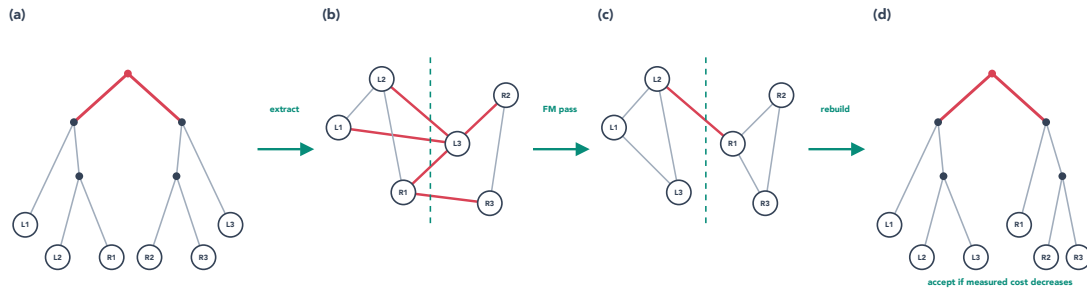


Figure 5: One surgery iteration on a six-tensor toy network (Algorithm 1, lines 3–10). (a) The most expensive tree node—the waist, red—divides the tensors into two subtrees. (b) *Extract*: the two leaf sets become a cut of the original network, with the straddling indices (red) carrying the cut weight. (c) *FM pass*: refinement moves tensors across the cut to reduce its cost (Fig. 2). (d) *Rebuild*: each side of the improved cut is ordered separately, the sides are joined, and the candidate tree is accepted only if its independently measured total cost decreases.

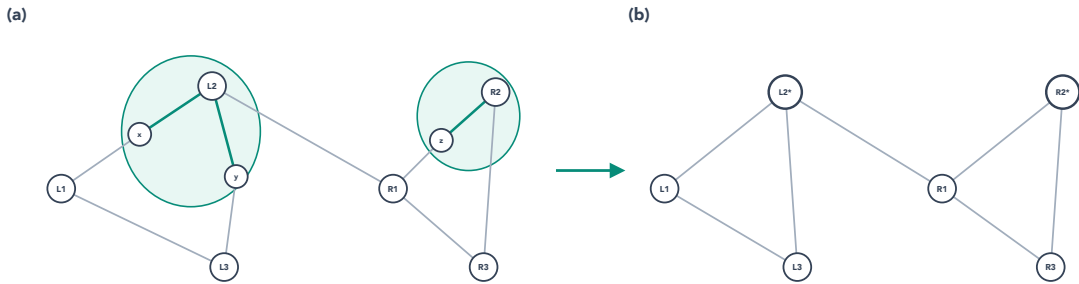


Figure 6: The exact simplification pass on a six-tensor toy network. (a) The pass greedily contracts adjacent (super-)tensors A, B whenever $\text{rank}(A \cdot B) \leq \max(\text{rank}(A), \text{rank}(B))$: here the shaded fragments absorb x, y into L_2 and z into R_2 . (b) The resulting super-tensors L_2^*, R_2^* form a smaller network for the order search; their internal contractions are retained and restored in the final tree as fixed subtrees.

the DBN instance).

6 Evaluation

Records. Figure 4b assembles the campaign’s confirmed records against the tuned reference on all ten instances. Global cut surgery holds three (king graph 36.96, surface code $d=21$ 47.40, reg3_1000 131.07—the only record on an expander in the entire campaign), the simplify-then-anneal line holds the real-Sycamore record (59.91), and the elimination seed holds the DBN record (29.00). On four instances—both certified closed instances, the 97-qubit random circuit, and 28-queens—the tuned reference itself remains unbeaten, exactly as the certified null of Appendix A predicts for the converged cases. A representative surgery run (Fig. 13, Appendix C) shows the dynamics: each accepted rebuild drops the incumbent below the pre-surgery record, and the official median lands at 47.377.

The time–quality Pareto front. Figure 7 presents the same-machine campaign as a Pareto scatter, following the presentation of the OMEinsumContractionOrders benchmark study [15]:

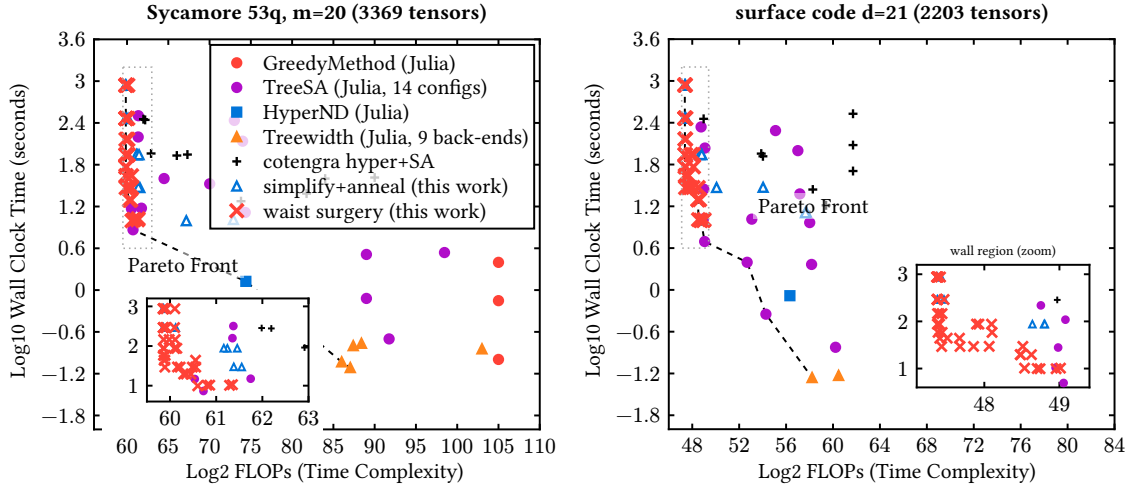


Figure 7: Time-versus-quality Pareto view of the same-machine campaign (dedicated 2-core server, all runs single-threaded), in the presentation style of the OMEinsumContractionOrders benchmark study [15]. Each point is one optimizer configuration or repetition. Baselines are the published implementations: Julia TreeSA across a 14-configuration ladder (both memory-target settings, 1–8 restarts, five sweep depths), GreedyMethod, HyperND, nine treewidth back-ends, and the cotengra hyper-optimizer with SA refinement at four budgets. Open blue triangles and red crosses are this work, at budgets from 10 to 900 s. Our tuned TreeSA port (not shown) coincides with the Julia TreeSA points wherever configurations overlap. The dashed line is the non-dominated front. Global cut surgery (crosses) holds the entire high-quality wall of the front on both instances; the insets zoom the wall region (dotted window), where the separation is explicit: on Sycamore the surgery front sits at $tc = 59.85\text{--}59.9$ from 60 s on (first reaching 59.89 at 30 s) while every TreeSA point stays at 60.5–61.8 at *all* budgets, and on surface code surgery reaches the 47.4 floor at 30 s while no TreeSA configuration gets below 48.75 at any measured budget. Orderings beyond the axis range (BFS/MCS-style treewidth, up to $tc = 443$) are omitted.

every optimizer configuration from every toolchain is one point, and the dashed line traces the non-dominated front. On both instances the front has the same anatomy: treewidth back-ends hold the fast/low-quality end (sub-second, 10–40 bits off best), HyperND the middle, and *global cut surgery holds the entire high-quality wall*—at every time tier (90/300/900 s) its points sit strictly left of every tuned-TreeSA point from either implementation. On the real Sycamore network surgery reaches 59.87 at 90 s while the tuned reference does not reach it even at 900 s (60.50)—a quality gap, not a speedup; on surface-code $d=21$, surgery at 90 s (47.39) matches the reference at 900 s (47.38)—a tenfold time-to-quality gain with both converging to the same floor.

Distributions and scaling. Figure 16 (Appendix C) gives the full run distributions. Over 15 repetitions at 90 s, the surgery and reference distributions on surface-code $d=21$ are fully separated—surgery’s worst run (47.91) beats the reference’s best (48.18); king-graph and reg3_1000 medians improve by 3.0 and 2.4; the elimination-seed mechanism shows a 14× tighter spread than the reference on the DBN instance. On the surface-code family the surgery advantage grows with distance (+0.04, +0.02, +0.11, +1.04 at $d = 9, 13, 17, 21$; Fig. 4c) while its run-to-run spread stays collapsed: global cut repair pays off precisely at scale, where

instance	this work	Julia OMECO suite		cotengra
	record (90 s)	best ≤ 90 s	best ≤ 900 s	SA, 90 s
sycamore_53_20_0	59.91	64.49	61.35	66.03
surfacecode_d21	47.40	48.98	48.75	54.03
ksg	36.96	39.22	38.27	69.04
reg3_1000	131.07	—	—	165.21
dbn_13	29.00	28.03	28.03	41.15
qft_27	29.61	29.62	29.62	31.45

Table 2: Records versus external baselines, all measured on the same server. Julia columns: over the configuration ladder of the reference implementation (OMEinsum-ContractionOrders.jl: TreeSA at $sc_{\text{target}} \in \{20, \infty\}$ with 1/4/8 restarts, HyperND, treewidth back-ends, run through its own harness), the best row whose measured runtime fits budget B . Cotengra column: median of three 90 s single-threaded runs of the hyper-optimizer followed by subtree simulated annealing; under this protocol it is starved of the parallel trial sampling that is its main strength, so these rows are a protocol-matched reference, not a claim about cotengra at large. Bold marks the winner per instance. The dbn_13 row is won by the treewidth back-end (min-fill), instantly—see Sec. 7. reg3_1000 is not in the Julia suite. nqueens_28 is excluded: no ordering is stable at these budgets (see text).

289 the annealer’s frozen waist is largest.

290 **External baselines.** Table 2 closes the loop against the reference Julia implementation, run
 291 on the same server through its own benchmark harness at a ladder of configurations, against
 292 the suite’s published numbers, and against cotengra. Our records beat the best same-machine
 293 Julia configuration *at ten times our budget* on the three large structured instances (Sycamore
 294 59.87 versus 61.35; surface code 47.40 versus 48.75; king graph 36.96 versus 38.27), and beat
 295 the entire published multi-optimizer suite by 6.8, 4.9, and 2.0 bits respectively. The cotengra
 296 column adds a protocol-matched cross-toolchain reference: on the small converged instances
 297 it confirms the shared frontier (reg3_250 40.5 versus our 39.9; the circuit proxy 62.4 versus
 298 61.5), while on the large instances a single-threaded 90-second budget starves its sampling-
 299 based portfolio (69.0 on the king graph, 187 on the 97-qubit circuit), which quantifies how
 300 much of that toolchain’s power lives in parallel trial throughput rather than in per-trial quality.

301 **Where the advantage ends.** Honest boundaries, all measured. On reg3_1000 the reference
 302 overtakes surgery at 900 s (129.8 versus 134.4): expanders have no cheap alternative sepa-
 303 rators for FM to find, so the surgery advantage there is a fixed-budget effect. On the DBN
 304 instance the published (and reproduced) treewidth row wins outright (28.03, instantly, versus
 305 our 29.00)—elimination methods are the right tool for dense hyperedge-structured networks,
 306 a theme Sec. 7 develops. On nqueens_28 no stable ordering exists at these budgets: our
 307 reference’s fifteen-run spread on the server spans 104.9–159.3, single Julia runs land inside
 308 that spread, and we decline to claim a direction. The Sycamore simplify-then-anneal pipeline,
 309 tuned against a faster development machine, is 0.5 *worse* than the reference at 90 s on the
 310 slower server and recovers only at 300 s—optimizer pipelines calibrate to absolute time, and
 311 cross-machine reporting must say so.

312 7 Application: inference workloads

313 Contraction ordering is not a quantum-only problem: exact probabilistic inference contracts
 314 the tensor network of a graphical model, and TensorInference.jl [21] does so literally, opti-
 315 mizing an einsum whose inputs are one unity tensor per variable plus one tensor per factor,
 316 at full cardinalities. The package’s own cross-tool evaluation [22] shows it competitive with
 317 dedicated inference engines (Merlin, libDAI, JunctionTrees) on the same networks, so its end-
 318 to-end runtime is bounded directly by the ordering quality studied here. We exported all 65 in-
 319 stances of the UAI-2014 MAR benchmark from the package’s own artifact in exactly that form,
 320 ranked them by a 5-second hardness scan, and measured the ten hardest (three dense deep-
 321 belief networks, four pedigree-linkage networks, one CSP, one grid, plus the package’s own
 322 benchmark instance Promedus_14) under the full protocol: our methods at five repetitions
 323 each, against a same-machine Julia ladder that includes the package’s two defaults (the signa-
 324 ture default GreedyMethod and the benchmark configuration TreeSA(ntrials=1, niters=5)),
 325 HyperND, the min-fill treewidth back-end, and tuned TreeSA.

326 Figure 8 shows every method’s distance above the per-instance frontier. Three findings.
 327 *First, the shipped defaults are 12–30 bits off frontier* ($4000\times$ to $10^9\times$ in flops): GreedyMethod
 328 lands 14.6–16.6 above on the DBN family and 30.5 above on linkage_15; the benchmark
 329 TreeSA configuration is little better there (10–16 above). Whatever mechanism one prefers,
 330 an inference user who accepts defaults pays orders of magnitude for it. *Second, the frontier*
 331 *splits into two structural regimes*. On dense DBNs, elimination methods win outright: min-fill
 332 treewidth and HyperND reach the frontier (25.9–28.0) in milliseconds, tuned annealing trails
 333 by 5–8 bits, and our elimination-seeded annealer (Sec. 5.2) lands within 0.7 of the elimination
 334 frontier with a $10\times$ tighter spread than the plain reference. On pedigree linkage the regimes
 335 invert: min-fill is 6–15 bits above the frontier, HyperND’s back-end (KaHyPar) segfaults on
 336 all four instances, and annealing-family methods—ours leading on all four—hold the fron-
 337 tier. *Third, the regime boundary is a structure property, not a size property*: the DBN networks
 338 have 44–66 shared labels of high degree (elimination’s sweet spot), while linkage networks
 339 are sparse with a thousand variables. The hard families are not a quirk of our selection: an
 340 early triage in the package’s own issue tracker found 105 of the 142 UAI-2014 instances fail-
 341 ing, and explicitly asked whether the contraction-order optimizer was leaving the DBN and
 342 pedigree families far from the treewidth optimum [23]. The reference tree decompositions
 343 posted there (Tamaki’s solver) let us answer that question with an external anchor: on all four
 344 dense instances in our hard set the best elimination-method space cost equals the reference
 345 treewidth exactly (sc = 21/20/22/20 against largest bags 22/21/23/21 on DBN_12/14/16
 346 and Grids_15), so the 12–30-bit default gap is measured against a certified-width frontier, not
 347 merely against the best order we happened to find. A portfolio that probes elimination first
 348 (milliseconds) and falls back to the annealing pipeline otherwise captures the frontier every-
 349 where we measured; the simplification pass (Sec. 5.1) additionally absorbs the per-variable
 350 unity tensors that inference-style einsums carry by construction.

351 **Beyond the benchmark: the excluded relational family.** The same issue tracker [23] ex-
 352 cludes one family outright—the five *relational* Markov-logic networks, “the problem scale is
 353 too large”—and no shipped test ever touches them. We exported and measured all five (Ta-
 354 ble 3); at 3 000–70 000 tensors they are up to $13\times$ larger than anything in Sec. 6, and they take
 355 the regime split of this section to its extreme. On the two dense instances the entire annealing
 356 family—our tuned reference, the Julia TreeSA ladder, and the shipped defaults alike—lands at
 357 *exactly twice* the optimal width (largest intermediate 2^{200} versus reference treewidth 100 on
 358 relational_4; 2^{500} , the full joint, versus 250 on relational_1) at every budget up to 900 s, while
 359 elimination orderings sit exactly on the Tamaki reference width in seconds. On relational_1

instance	tensors	labels	ref. width	annealing family (best)		elimination (Julia MF)
				either toolchain	this work + VE seed	
relational_1	63 000	500	250	503.5 (sc 500)	260.3 (sc 250)	259.5
relational_2	38 750	13 000	2	17.44	17.44	17.44
relational_3	3 000	1 000	7	16.77	16.78	16.93
relational_4	30 400	10 200	100	202.9 (sc 200)	109.0 (sc 100)	108.2
relational_5	70 000	10 000	10	23.93 ($\sim 10^3$ s)	24.03 (90 s)	23.93 (29 s)

Table 3: The relational family, same server. “Ref. width” is the Tamaki reference treewidth from [23]. “Annealing family (best)” is the best of our tuned reference, all attempts of Secs. 5–6, and the full Julia TreeSA ladder, at budgets up to 900 s (relational_1: no method of our toolchain emitted any order before the seed of the final row existed; the value shown is Julia’s default greedy after 433 s). “This work + VE seed” is the pipeline seeded by the scalable quotient-graph elimination order (median over repetitions; identical across all budgets 90–900 s). Elimination reaches the reference width exactly on every instance; the dense pair is decided by ~ 94 and ~ 244 bits.

our own toolchain initially produced *no order at all* within any budget: the greedy construction stage was quadratic in practice and never completed—a defect this campaign exposed and we have since fixed in the library. Surgery, instrumented on relational_4, maps its own boundary honestly: on some seeds bounded FM finds enormously cheaper cuts (36–43 bits) and surgery is the only refiner to escape the plateau (190.5 at 90 s), but the improved separators are near-cliques (clique fraction 0.41–0.44, against 0.01–0.05 on the instances of Sec. 4), and on other seeds the waist is locally minimal on every call—balanced bipartitions are structurally wrong for hub-dominated networks, whose optimal trees are deeply unbalanced elimination peelings outside the balance class surgery searches.

Answering in kind: a seed that scales. The portfolio rule above is only as good as its elimination probe, and ours froze beyond $\sim 4\,000$ labels. Diagnosing the freeze showed it was largely an implementation artifact (the quadratic greedy blocked the pipeline before the seed ran), and the cure is the standard machinery of sparse-matrix ordering: a quotient-graph elimination engine with element absorption and lazy weighted-min-degree scoring [24] orders relational_4 in 72 ms at width 101—the optimum—and relational_1 in 1.9 s at width 251. Seeded with it, our pipeline matches the elimination frontier on every relational instance (Table 3), reaches relational_5’s width-10 floor at 90 s where the annealing ladder needs $\sim 1\,000$ s, and puts the first number of any method in this study on relational_1. The engine is released as a first-class Treewidth optimizer in the omeco library, alongside the greedy scaling fix.

8 Conclusions

We introduced global cut surgery, a contraction-order optimizer built on a structural measurement: the converged annealer’s dominant contraction—the cut that carries its cost—is never a locally minimal balanced cut of the tensor graph (835 instrumented calls, zero exceptions, margins of 3–12 bits). Surgery extracts that cut, improves it globally with bounded Fiduccia–Mattheyses passes, cold-rebuilds both sides with fixed-work sub-anneals, and accepts on measured cost. Together with an exact simplification preprocessing pass, it sets records on real quantum-circuit, surface-code, and counting networks; separates fully from tuned annealing’s

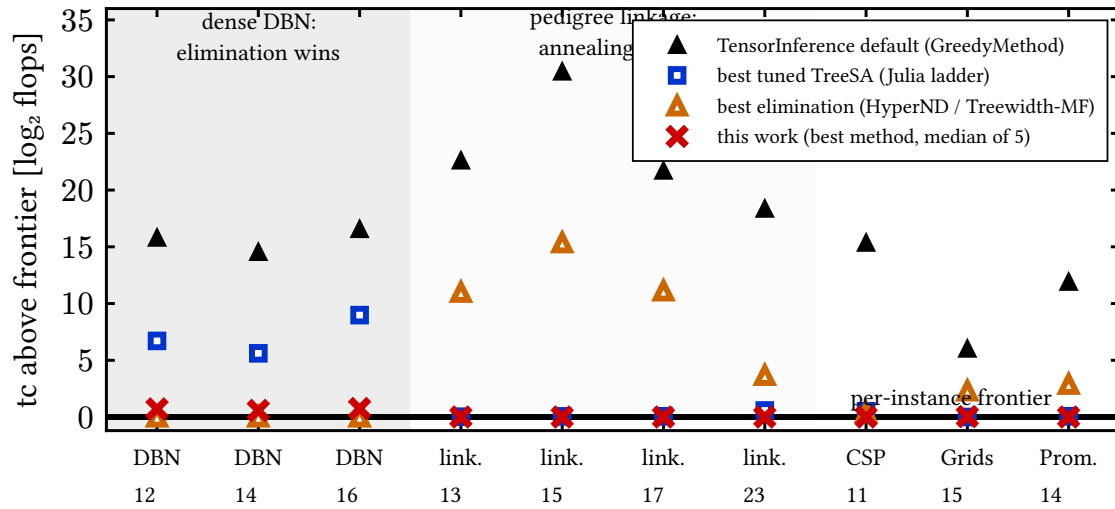


Figure 8: The ten hardest UAI-2014 MAR instances (from a 65-instance scan): each method’s tc above the per-instance frontier, same machine, 90 s budgets for search methods. Filled triangles: TensorInference.jl’s signature default. Squares: best tuned TreeSA from the Julia ladder. Open triangles: best elimination method (min-fill treewidth / HyperND; KaHyPar segfaults on all four linkage instances, so its rows there are min-fill only). Crosses: this work (best method, median of five). Shaded bands mark the two regimes.

run distribution; holds the entire high-quality wall of the same-machine time-quality Pareto front against the reference Julia implementation, its published suite, and cotengra; and its advantage grows with problem scale.

The algorithm’s design is uniquely constrained by negative results: on converged instances the annealing frontier is certifiably width-optimal (optimum in $[53, 61.5]$ on the circuit proxy, with profile-based bounds provably exhausted), and twenty falsified mechanisms show that global information helps only as input, never as search control. The method’s boundaries are mapped: expanders return the advantage at long budgets, dense inference networks belong to elimination orderings (our elimination-seeded variant comes within 0.7 bits), sparse pedigree networks belong to annealing—and shipped inference defaults sit 12–30 bits above either regime’s frontier, so a milliseconds-cheap elimination probe belongs in front of every annealer.

Deliverables: the surgery pass, the simplification pass, tuned defaults, a warm-start annealing API, a scalable Treewidth elimination-order optimizer, a fix for a quadratic scaling defect in greedy construction, and two further bug fixes are released in the omeco library (prepared for upstreaming to OMEinsumContractionOrders.jl), together with the validator, all certificates, and the 58-attempt audit trail. Open problems: carving-width-type lower bounds; the two instances that repelled every mechanism (the 97-qubit random circuit and 28-queens, the latter unstable at practical budgets); heterogeneous bond dimensions; sliced contraction, where the interaction between slicing and surgery is unexplored; and *unbalanced* surgery—the relational family shows the balance prior excludes the elimination-like separations hub-dominated networks need, so a cut repair that searches across balance classes (or peels rather than bisects) is the natural next mechanism.

Acknowledgements

The author thanks the OMEinsumContractionOrders.jl and TensorInference.jl communities for the open benchmark suites and artifacts this study builds on.

Code and data availability The optimizer library (omeco), the validation harness, all certificates (treewidth minors, cut sets, and their verification scripts), per-attempt logs, cycle reports, the same-machine campaign data (305 + 200 + 80 + 69 + 30 scored runs, the 108-run surgery budget grid, the 324-run relational campaign, and the Julia ladders), and the UAI instance exporter are available in the project repository. Every figure regenerates from JSON data files via Typst sources shipped with the manuscript.

Use of AI systems This research was conducted as a steered autonomous research campaign: LLM-based agents proposed hypotheses, implemented attempts in isolated worktrees, and drafted analyses, under human steering decisions documented verbatim in Appendix E and under a validation protocol in which no agent-reported number is trusted—every scored quantity is recomputed by an independent checker from artifacts, and all certified claims are verified by machine-checkable certificates independent of any agent output. The manuscript was drafted with the same agents and revised by the human author, who takes full responsibility for its content.

Funding information To be completed before submission.

A Why search control cannot win: a certified null

The case for operating on the input rather than the search is made by a null result, and we certify it. On the two closed instances the tuned annealer converges within the budget, and nothing we or an independent toolchain tried can beat it—for reasons that are provable, not anecdotal.

A.1 Thirteen mechanisms, two toolchains, one frontier

We attempted to beat the tuned reference with mechanically distinct search paradigms, implemented independently and scored under the protocol of Sec. 2: coarse subtree-transfer moves and basin hopping [17], evolutionary replica populations, exact-penalty ramps, tabu search with strategic oscillation, large-neighborhood destroy-and-repair, beam search with lookahead, Boltzmann-greedy portfolios [4, 25], recursive balanced min-cut construction [5, 26], elimination-order annealing [6], order-then-tree interval dynamic programming [27], hierarchical coarsen-then-exact solves, structure-guided sweeps, and profile-aware annealing.

Figure 9 shows every scored result as its distance from the best verified value. All thirteen mechanisms land within $0.35 \log_2$ of the tuned reference on both instances—most within noise—and none set a confirmed record. An independent toolchain reproduces the same frontier: cotengra’s simulated-annealing refiner [4] ties it to within 0.05 and 0.14, while cotengra’s hyper-optimization mode alone remains 1.7 and 2.4 behind at matched budget. Extending the budget tenfold moves the frontier by less than 0.1. At this scale the frontier is a property of the cost landscape, not of any search mechanism, schedule, or implementation.

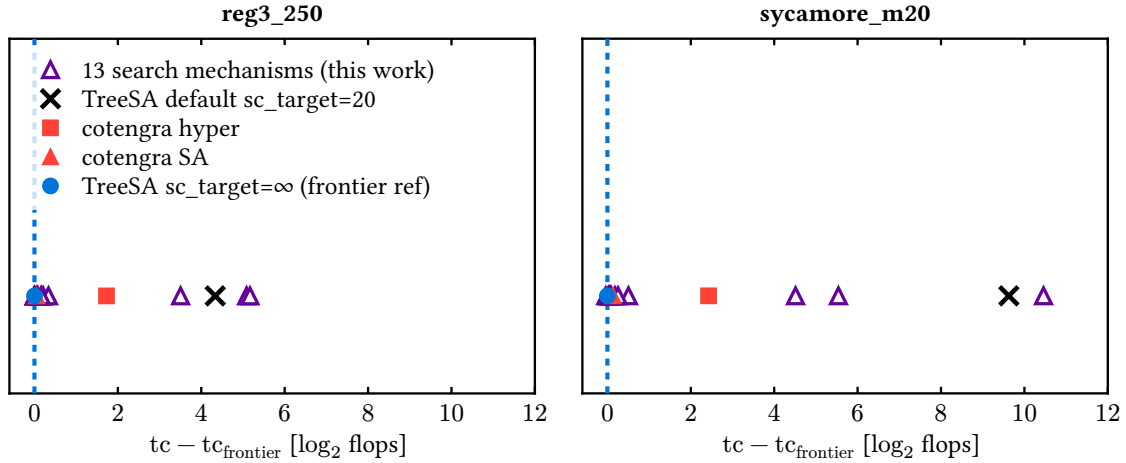


Figure 9: Distance of every scored optimizer from the frontier value (dashed line at zero). Open triangles: the thirteen search mechanisms of this work. Filled symbols: the tuned TreeSA reference ($sc_{\text{target}} = \infty$, circle), cotengra’s SA refiner (triangle), cotengra’s hyper-optimizer (square), and the shipped-default TreeSA (cross).

447 A.2 Certified bounds: width-optimality and an impossibility result

448 Every classical lower-bound technique for contraction cost is *max-form*: it bounds the single
 449 largest contraction. The Markov–Shi correspondence [1] equates the minimal largest contrac-
 450 tion rank with the treewidth of the line graph $L(G)$, and any balanced edge of a contraction
 451 tree yields a cut whose boundary lower-bounds the largest contraction.

452 On sycamore_m20 these tools give a sharp result. An explicit temporal cut severing all
 453 53 qubit world-lines gives a balanced cut of value 53; extensive search finds nothing smaller,
 454 and the frontier trees achieve $sc = 53$ exactly. The frontier is therefore *width-optimal*, and
 455 $tw(L(G)) = 52$. Combined with the frontier value, we obtain the certified localization

$$tc_{\min}(\text{sycamore_m20}) \in [53, 61.544]. \quad (\text{A.1})$$

456 On reg3_250 the analogous certification is obstructed by the known hardness of certifying
 457 high treewidth on expanders: the best machine-verified certificate (a 32-vertex minor of $L(G)$,
 458 checked by two independent exact solvers) gives $tw(L(G)) \geq 18$ against a constructive upper
 459 bound of 34.

460 The width bound has a ceiling: tc is a log-sum (1) over $\sim |V|$ contractions, and the fron-
 461 tier’s excess over the width bound—8.5 on sycamore_m20, close to the maximal possible
 462 $\log_2 561 = 9.1$ —is invisible to any bound on a single contraction. Closing the interval (A.1)
 463 from below requires *sum-form* bounds. For any tree, each internal node v with leaf set S_v
 464 satisfies $\text{cost}_v \geq |\partial S_v|$, the boundary of S_v in G . Writing $b(k)$ for the isoperimetric profile of G
 465 (the minimal boundary over all k -vertex subsets), we prove:

466 **Theorem 1** (Sum-form profile bound). *For every contraction tree, $tc \geq \log_2 F(n)$ where F obeys*
 467 *the recursion $F(1) = 0$, $F(k) = 2^{b(k)} + \min_{1 \leq j \leq k/2} [F(j) + F(k-j)]$.*

468 **Theorem 2** (Dyadic-window bound). *Descending from the root always into the larger child*
 469 *visits, for every dyadic window $W_j = (n/2^{j+1}, n/2^j]$, at least one internal node whose leaf-set size*
 470 *lies in W_j ; these nodes are distinct, hence $tc \geq \log_2 \sum_j 2^{\min_{k \in W_j} b(k)}$.*

471 Both theorems are validated computationally (exhaustively at small n , and on 3000 ran-
 472 dom trees). Both fail to beat the max-form cap (Fig. 10): on sycamore_m20 they reach 47.2
 473 and 47.0 against the carving value 53. The reason is structural:

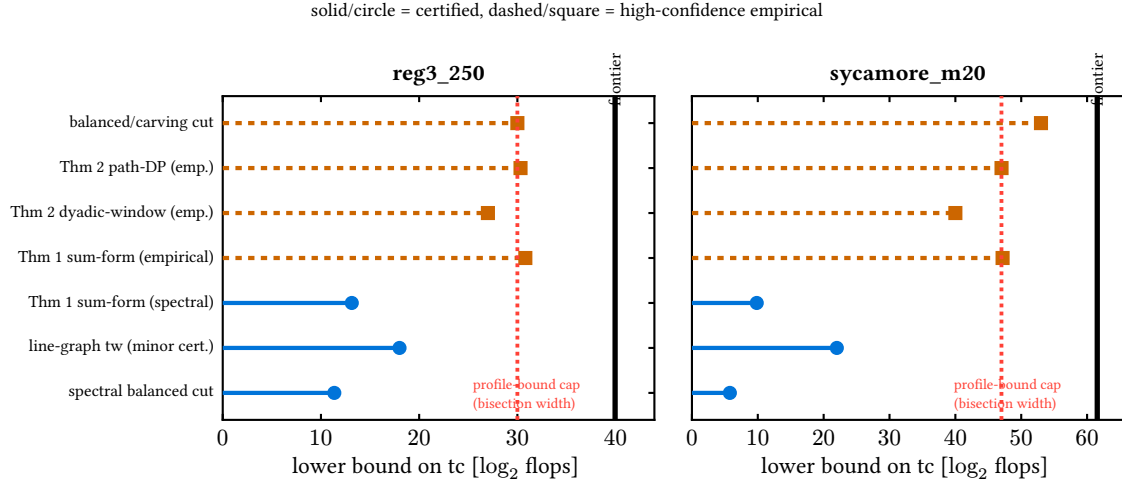


Figure 10: Lower-bound ladder for both closed instances. Solid stems with circles are certified (machine-checkable certificates); dashed stems with squares are high-confidence empirical values. Every profile-based bound (Theorems 1 and 2) is capped at the bisection width (dotted line, Observation 1); only the balanced/carving cut exceeds it. The solid vertical line is the optimizer frontier.

Observation 1 (Profile bounds are capped at the bisection width). *Any lower bound computed from the isoperimetric profile $b(\cdot)$ alone is at most $\max_k b(k) + \log_2 \log_2 n$, and $\max_k b(k)$ is the bisection width—strictly below the balanced-cut (carving) value whenever the profile dips off-center. On sycamore_m20 it does: we exhibit a triply-verified 141-tensor subset with boundary exactly 40, against temporal-slab boundaries of 53 (Fig. 11).*

The $47 \rightarrow 53$ gap lives in jointly-nested-cut structure that per-size minima discard; the $53 \rightarrow 61.5$ gap is the log-sum residual. A complementary measurement (Fig. 12) shows why search cannot exploit the residual either: reducing the largest contraction by one unit systematically multiplies the number of contractions just below the peak, so total flops behave as if conserved under profile reshaping. This certified null fixes the design space for any challenger: the gains must come from changing what is searched, not how.

B What does not work: falsified alternatives

Twenty further mechanisms were implemented, measured, and falsified under the same protocol, and their taxonomy (Table 4) is what makes the design of Sec. 5 non-arbitrary: *every mechanism that injected global information as search control failed; every mechanism that injected it as input won.* Construction-time global structure (separators, spectral cuts, boundary combs) loses to the greedy portfolio it seeds; search-control globality (targeting, tempering, blocking, acceptance relaxation) breaks the annealer’s schedule invariants; exhaustive enumeration hits certified combinatorial walls (window-exact optimality of the incumbent frontier; block counts exploding past 10^5 within bits of the minimum boundary—with an outer-product counterexample showing the connectivity-restricted variant is not even correct); and population policies drain a small recombination reservoir (about ten productive recombinations) regardless of diversity management. Global cut surgery is, to our knowledge, the unique surviving design point: it consumes global information (a better cut) as an *input* to a rebuild, leaving the annealer’s control loop untouched.

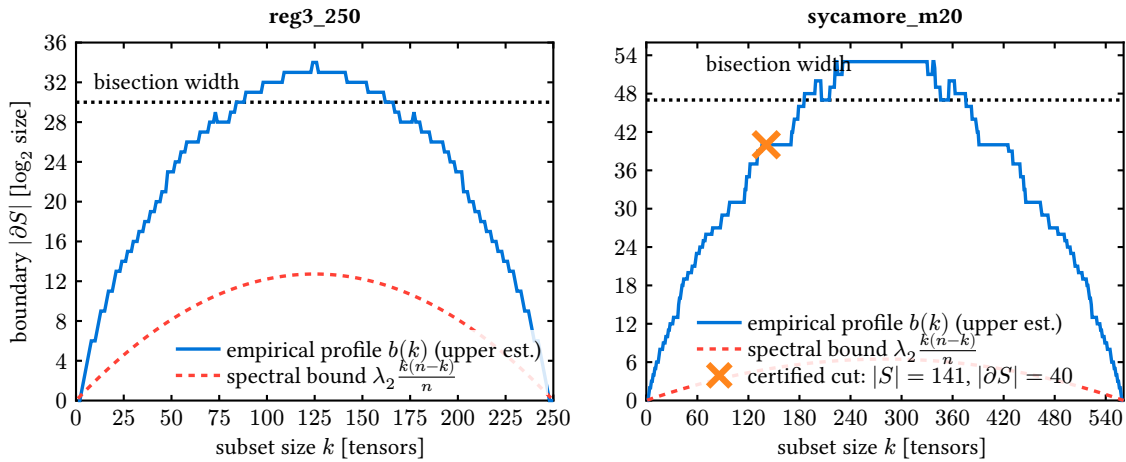


Figure 11: Isoperimetric profiles. The empirical profile $b(k)$ (solid) dips well below the bisection width (dotted) at off-center sizes; the certified marker shows the 141-tensor sycamore_m20 subset with boundary exactly 40. This dip is what caps all profile-based bounds (Observation 1).

499 C Supplementary figures

500 D Methodology: a research loop that distrusts itself

501 The results above were produced by an autonomous research loop—56 optimizer attempts
 502 over ten cycles, each in an isolated worktree with a pre-registered hypothesis, scored by a
 503 sandboxed validator—and the loop’s central design lesson is that its most important outputs
 504 were *retractions* (Fig. 17). Three failure modes were caught by the harness rather than by
 505 luck:

506 (i) *Schedule exploitation*. The first gate (a speed-versus-baseline bar) was passed by budget
 507 hygiene alone—anytime writes and early termination—with no algorithmic content. The fix
 508 was structural: fold the exploit into the reference, making schedule-only candidates score zero
 509 by construction, and keep the exploit as a permanent negative control.

510 (ii) *Mis-tuned-baseline gains*. Eight mechanisms then “beat” the references by up to $800\times$ —
 511 until a strengthened reference (the one-line sc_{target} fix of Sec. 3) reclaimed the entire gain.
 512 What survived was the tuning discovery itself, and the leaderboard practice of ratcheting refer-
 513 ences to absorb every discovered improvement.

514 (iii) *Instrument bias*. A failed attempt’s diagnosis revealed that the independent scorer
 515 silently rejected contraction trees deeper than ~ 475 (a recursion limit), biasing the admissible
 516 space against exactly the deep, MPS-like trees that suit circuit instances. The scorer was fixed,
 517 its negative-control suite re-passed, and the affected hypothesis re-tested.

518 As the campaign scaled, the harness grew with it: parallel scoring lanes with per-child CPU
 519 accounting, median-of-three scoring above 1000 tensors, early abort of hopeless confirmation
 520 runs, and harness-clocked anytime curves that make time-to-frontier claims backdate-proof.
 521 The pattern that generalizes: score only through a hardened independent validator; maintain
 522 executable negative controls, and when a new exploit is found, patch the harness and add the
 523 exploit to the control suite; ratchet references so yesterday’s wins are tomorrow’s baselines;
 524 and report distributions wherever single runs are noisy. Under this regime, every claim in this
 525 paper is either machine-certified, or labeled high-confidence with its evidence, or was retracted
 526 before publication—and the audit trail (per-attempt logs, certificates, and cycle reports) ships
 527 with the code.

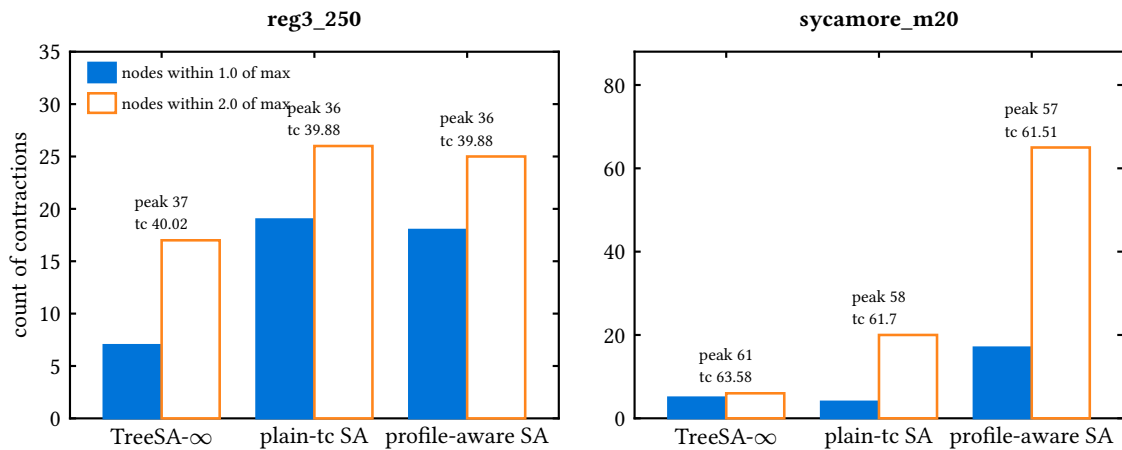


Figure 12: Cost-profile conservation at the frontier. Bars count contractions within 1.0 (solid) and 2.0 (outline) of the largest; labels give each tree’s peak cost and tc. Trees that lower the peak by one unit acquire several times more near-peak contractions at essentially unchanged tc.

528 E How the algorithm was found: steered autonomous research

529 The algorithm of this paper was produced by an autonomous research loop (machine-proposed
 530 hypotheses, isolated implementations, hardened scoring) that was *steered* by a small number
 531 of human decisions. (For LLM-driven *heuristic evolution* on this same problem—automated
 532 search over code rather than a steered hypothesis loop—see [10].) Because the loop retained
 533 every artifact, the causal path from the opening question to the algorithm can be reconstructed
 534 exactly. Figure 18 shows that path left to right: the top row quotes the steering prompts, the
 535 middle row shows only the attempts that carried results forward (most of the 56 attempts
 536 served as controls, falsifications, or dead branches feeding Appendices B and D), and each
 537 arrow states why the transition happened.

538 Three properties of this trajectory are worth stating explicitly. *First, every pivot was forced by*
 539 *a measurement, not chosen freely.* The move from tc-optimization to the anytime axis happened
 540 because thirteen mechanisms and a second toolchain had converged on one frontier that was
 541 then certified locally optimal three independent ways; the move to scale happened because
 542 time-to-frontier proved heavy-tailed at the claim threshold; the move to global-information
 543 mechanisms happened because every search-control variant had failed with a structural cause
 544 while every input-level variant had won. *Second, the human contributions were few but load-*
 545 *bearing:* the demand for a strengthened reference (which produced the tuning result of Sec. 3),
 546 the switch to real-provenance instances (which unlocked the simplification pass—the same
 547 mechanism had been correctly measured as useless on the pre-simplified proxy two cycles
 548 earlier), the question “can global information help the local search escape?” (which became
 549 global cut surgery), and the directive to demonstrate the methods on inference workloads
 550 (which produced the two-regime result of Sec. 7). *Third, retests are cheap when provenance*
 551 *changes:* attempt 013’s null on the synthetic proxy and attempt 039’s records on the real circuit
 552 are the same mechanism measured on different inputs, and only the pre-registered instance
 553 provenance made the contradiction interpretable rather than embarrassing.

family / mechanism (attempt)	measured cause of failure
<i>Construction proxies</i>	
BFS/planar separators (037)	seeds lose to greedy portfolio even with separator injection
spectral bisection (041)	power iteration cannot converge (gap $\sim 1/n^2$); a perfect bisection is still dominated
boundary-MPS combs (036/044)	cutwidth 90–126 lower-bounds any comb, far above balanced-tree records
mass randomized greedy (040)	quality ceiling 15–208 bits above frontier at 1.2 builds/s
<i>Search-control globality</i>	
elimination-order (045)	annealing flat landscape: 30 improvements per 2.5M moves; representation floor $\sim 2\times$ above tree space
congestion-softmax (042)	targeting collapses onto the dominant node on peaky profiles; 40–80% dead traversal
cache-blocked regional SA (043)	pointer trees have no memory contiguity; blocking starves global moves
replica-exchange tempering (048)	swap acceptance $\rightarrow 0\%$ at scale (energy gaps grow with n)
late-acceptance hill climbing (051)	never cools on plateau-dominated landscapes; schedules are load-bearing
MCTS over prefixes (050)	reaches depth 3 of ~ 380 at 8–25 rollouts/s
<i>Exhaustive enumeration</i>	
window-exact splice (033)	frontier trees are window-exact optimal (≤ 16 leaves); the dominant window spans a constant fraction
Tamaki-style block DP (056)	feasible low-boundary blocks explode past 10^5 – 10^6 within bits of $\min_w$; outer-product counterexample falsifies the connectivity-restricted variant
<i>Population policy</i>	
path-relinking / consensus / diversity pools (046/049/055)	basin complementarity drains after ~ 10 recombinations regardless of pool policy

Table 4: The falsified-mechanism taxonomy. Each row is an implemented, scored, and diagnosed attempt; attempt numbers index the public audit trail. The four families fail for four distinct measured reasons, and none of the failures is a tuning accident.

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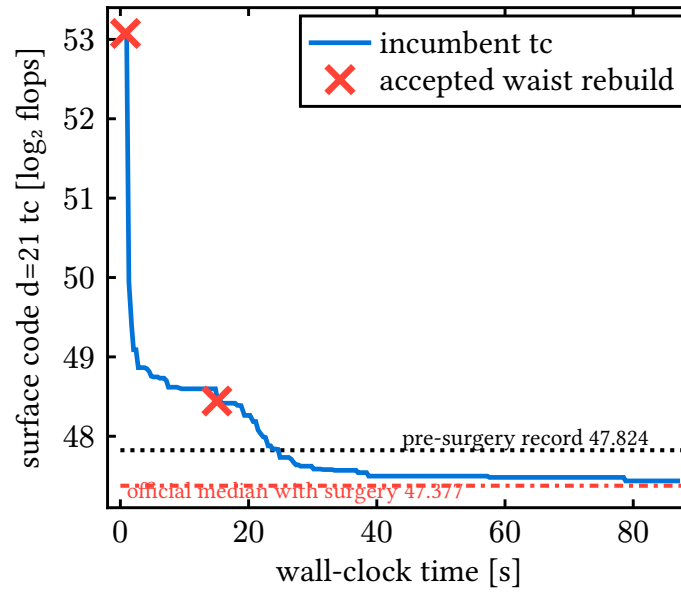


Figure 13: A representative surgery run on surface code $d=21$: incumbent tc (solid), accepted waist rebuilds (crosses), the pre-surgery record (dotted) and the official with-surgery median (dash-dotted). Each accepted rebuild moves the waist; surgery repeats at the new waist until no improving cut remains.

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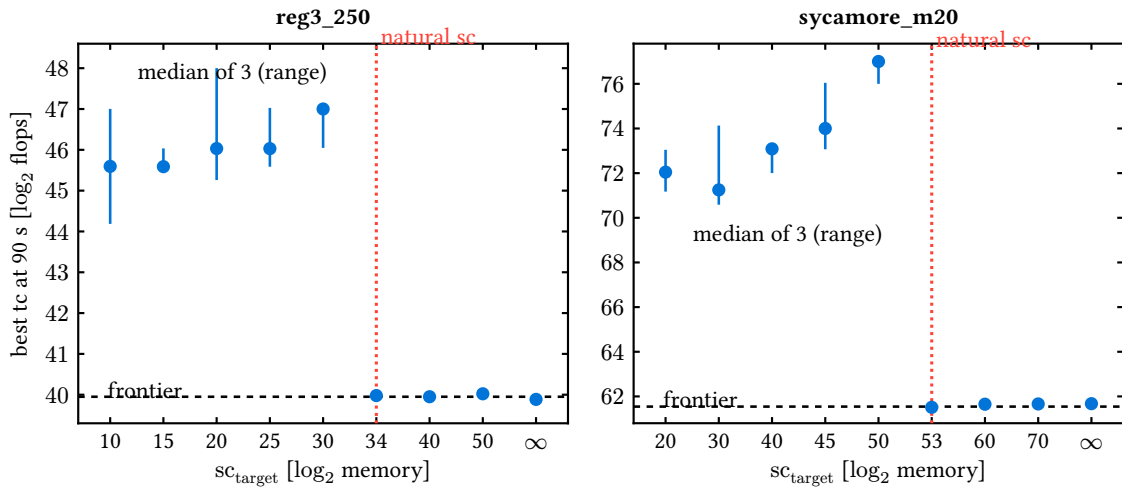


Figure 14: Best tc reached in 90 s of single-threaded annealing versus the memory target sc_{target} (median of three runs; whiskers span the range). The dashed line marks the converged frontier of Appendix A; the dotted vertical line marks the natural space cost of optimum-quality trees. The transition is a cliff at the natural value: every smaller target is comparably harmful, and every target at or above it, including $sc_{\text{target}} = \infty$, reaches the frontier. The shipped default is $sc_{\text{target}} = 20$.

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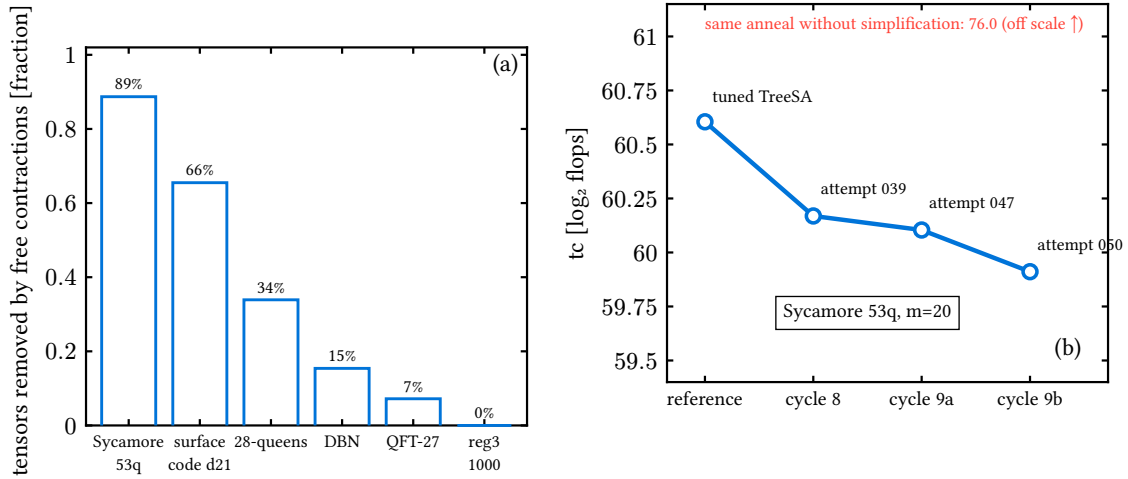


Figure 15: The preprocessing pass. (a) Fraction of tensors removed by the exact rank-non-increasing pass; the random-regular control is untouched at 0%. (b) The Sycamore 53q record march: the tuned reference (leftmost) and three successive simplify-then-anneal attempts. The same anneal without simplification reaches 76.0 (off scale): preprocessing, not search, carries this improvement.

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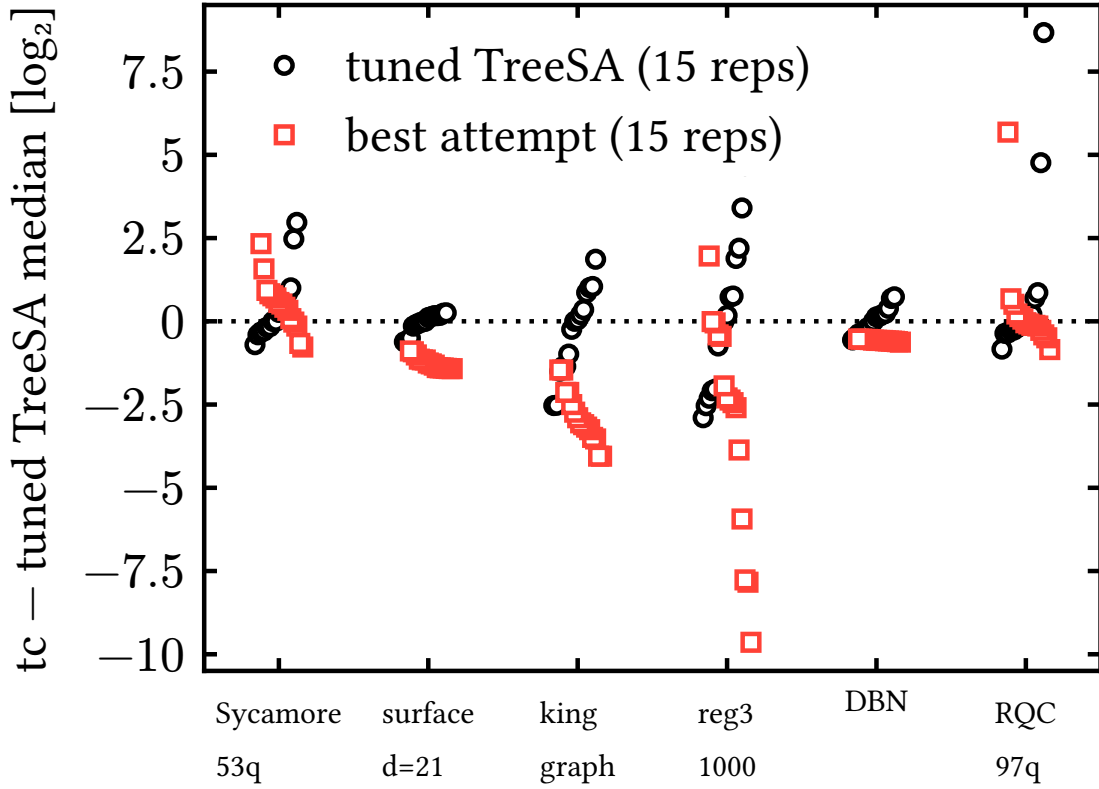


Figure 16: Run distributions of the same-machine campaign: all 15 repetitions per instance at 90 s, centered on the tuned reference's median. Circles are the tuned reference, squares the best method per instance.

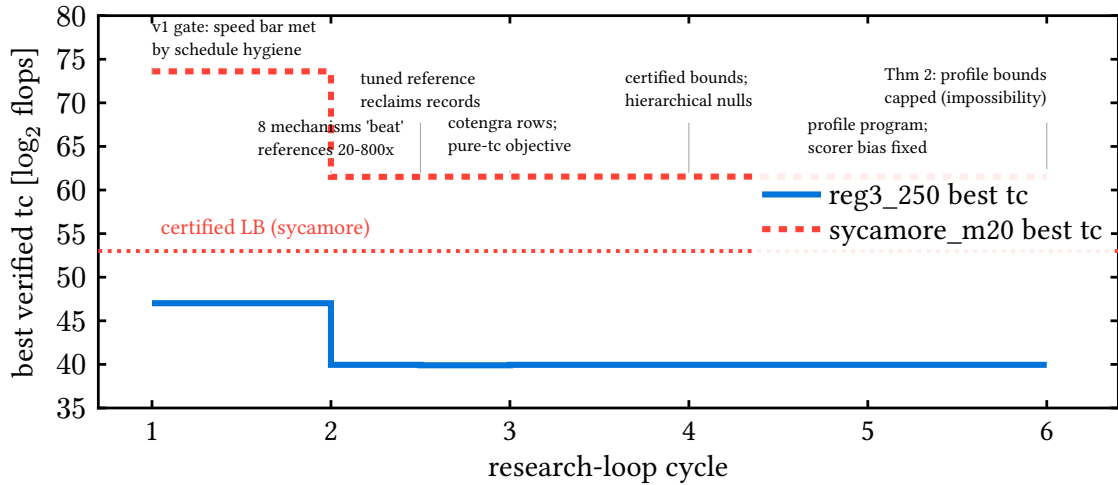


Figure 17: Best verified tc over the research-loop cycles. Apparent progress in cycle 2 is reclaimed by the strengthened reference (cycle 2.5); thereafter the closed-instance frontier is flat while the certified picture grows, and cycles 7–10 move the record board on the large structured instances instead. Annotations mark the three instrument-caught failure modes.

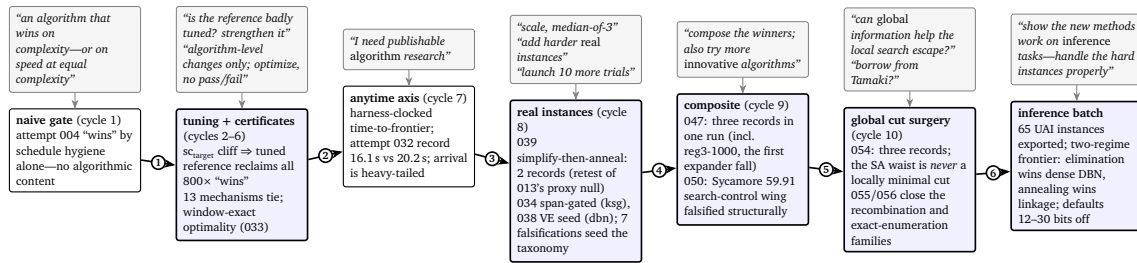


Figure 18: The research roadmap, left to right. Top row (grey): the human steering prompts, essentially verbatim. Bottom row: the result-carrying attempts only; most of the 56 attempts are controls and falsifications that feed Table 4 and Appendix D. Each numbered arrow is a transition forced by a measurement: (1) the gate was passed without an algorithm, so the gate was hardened and references ratcheted; (2) tc saturated and was certified locally optimal, so the objective axis changed to arrival time; (3) speed gains sat at the confirmation noise floor while scale budgets were non-converged, so the regime changed; (4) the winning mechanisms proved complementary, so they were composed while exploration continued; (5) every win injected global information as *input* while every search-control variant failed, so global information was aimed directly at the local search's blind spot—the dominant cut; (6) the mechanism portfolio was complete, so it was pointed at a second application domain to test transfer.